

Add as member functions to the class BST:

In Exercise 12.4 (page 697),

Problem 22, 24 and 25, see below:

22. Write a recursive function member `level()` for class template BST that determines the *level* in the BST at which a specified item is located. The root of the BST is at level 0, its children are at level 1, and so on.
23. Proceed as in Exercise 23, but write a nonrecursive function.
24. The worst-case number of comparisons in searching a BST is equal to its *height*, that is, the number of levels in the tree. Write a recursive function member `height()` for class template BST to determine the height of the BST.
25. Write a recursive function member `leafCount()` for class template BST to count the leaves in a binary tree. (*Hint*: How is the number of leaves in the entire tree related to the number of leaves in the left and right subtrees of the root?)

in addition to above add following functions

- `Size()` // should return the size(number of the nodes in a BST),
- `MirrorImage()` // should return another binary tree that is mirror image of the original,
- `isBST()` // should return value 1 if it is a binary tree or return false if it is not.
- Copy Constructor